

Lecture 2 (13.10.2025)

2's complement, overflow

CBW = convert byte to word

ADC

MUL unsigned

IMUL signed

2's complement:

→ representation of a negative number, $2^m - V$

→ "we interpret representations and represent interpretations"

1001 0011₍₂₎ = 93₍₁₆₎ = 147₍₁₀₎ unsigned interpretation

signed interpretation

1001 0011 → 8 bits → subtract from $\underbrace{1000000000}_{8 \text{ zeroes}}$

$$\begin{array}{r} 1000000000 \\ - 10010011 \\ \hline \end{array}$$

0110 1101₍₂₎ → the signed inter.

"

6D₍₁₆₎

"

109₍₁₀₎

2's complement of 1001 0011: 0110 1101

signed interpretation:

-109

11b (binary)

0F6h / 0xF6 (hexa)

10d

2's complement of 1001 0011:

↳ 1's complement: 0110 1100 + 1 $\Rightarrow$ 2's: 0110 1101.

2's comp. of 1001 0011: 0110 1101.

$m = 8$ bits.

unsigned: $[0, 2^m - 1] = [0, 2^8 - 1] = [0, 255]$

signed: $[-2^{m-1}, 2^{m-1} - 1] = [-128, 127]$

1001 0011b = 93h = 147d $\rightarrow$ unsigned 147.

= -109 $\rightarrow$ signed

$\Rightarrow$ 147 and -109 are complementary

1001 0011b = 147d or -109

Base 2, most significant bit 0.

range: $m = 8: [0, 255] \cap [-128, 127]$

if the number is positive: $[0, 127]$

Base 2, most significant bit 1.

Binary nr.

a) rep. of $\overline{0xxx}$ having the value +abc in unsigned int.; the value of $\overline{0xxx}$ in signed int.?

- the same -

b)

c)

d)

2's comp.

binary format	interpretation	value
<u>0xxx</u>	unsigned	+abc
	signed	+abc
<u>1xxx</u>	unsigned	+abc
	signed	2's comp

byte = 8 bits

word = 16 bits

dword = 32 bits

MOV dest, source

MOV al, 2

$1001\ 0011_{(2)} = 147_{(10)}$ unsigned

$-147_{(10)} = ?_{(2)}$

MOV dest, source ; $D \leftarrow S$.

MOV AL, 2 ; $AL \leftarrow 2$

MOV AX, AL ; assembler error

↓ ↓
16 bit 1 byte

16 bit 8 bit = 1 byte

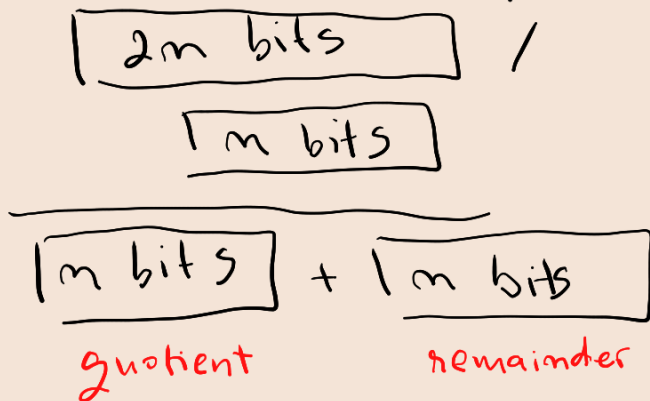
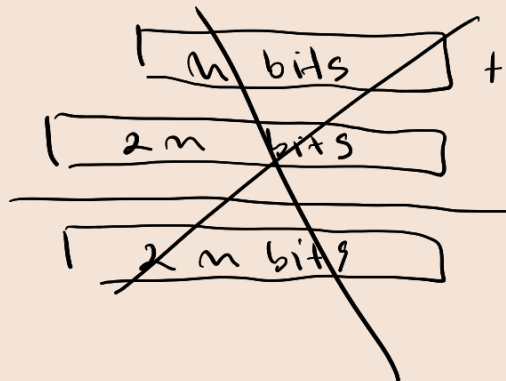
ADD dest, source

SUB dest, source

ADC ?

MUL op → op reg 8/16/32 bits
(unsigned) AL AX EAX

IMUL op
(signed)



Overflow:

CF, OF



Carry
flag

(unsigned)



overflow
flag

(signed)

147 + 116 = 262

>256

$$\begin{array}{r} 1001 \ 0011 + \\ 0111 \ 0011 \\ \hline 1 \ 0000 \ 0110 \\ \downarrow \\ CF=1 \end{array}$$

Overflow = mathematical situation which expresses the fact that the result of the operation did not fit the reserved space.

eg: $-147 \notin [-128, 127]$, $n=8$

We need $n=16$ or bigger.

Overflow (asm language): a situation that says that the last performed operation did not fit the reserved space.

or

does not belong to the admissible repres. interval used

or

- the repres. is a math nonsense.

CF vs OF.

$$\begin{array}{r} 1001 \ 0011 + \\ 1011 \ 0011 \\ \hline 1 \ 0100 \ 010 \\ \downarrow \\ OF=1. \end{array}$$

CF=1.

$$\begin{array}{r} -109 + \\ -77 \\ \hline \end{array}$$

$$\begin{array}{r} 147 + \\ 479 \\ \hline \end{array}$$

$$\begin{array}{r} 937 + \\ 834 \\ \hline \end{array}$$

$$\begin{array}{r} \text{---} \\ -186 \\ \hline \end{array}$$

signed

$$\begin{array}{r} 326 \quad 146h \\ \hline \end{array}$$

unsigned

Explanation: $326 \notin [0, 255] \Rightarrow [0, 511]$
 $m = 8 \text{ bits}$ $m = 9 \text{ bits}$

$$\begin{array}{r} 147 + \\ 179 \\ \hline + 70 \\ \hline \end{array}$$

on 8 bits
unsigned

$$\begin{array}{r} -109 + \\ -77 \\ \hline + 70 \\ \hline \end{array}$$

$m = 8 \text{ bits}$

$$\begin{array}{r} 0101 \ 0011 + \\ 0111 \ 0011 \\ \hline 1100 \ 0110 \end{array}$$

$$\begin{array}{r} 83 + \\ 115 \\ \hline 198 \end{array}$$

$$\begin{array}{r} 53h + \\ 73h \\ \hline 06h \end{array}$$

unsigned operands

$$\begin{array}{r} 83 + \\ 115 \\ \hline 198 \end{array}$$

OF = 1

$m = 8 \text{ bits}$

signed operands

MUL CF = 0

ADD $m = 8$

$[0, 255]$

$$\begin{array}{r} 0 \dots + \\ 0 \dots \\ \hline 1 \dots \end{array}$$

I cannot add
+ with + and get -.

$$\begin{array}{r} 1 \dots + \\ 1 \dots \\ \hline 0 \dots \end{array}$$

I cannot add
- with - and get +.

signed overflow OF.

$$\begin{array}{r} 1 \dots - \\ 0 \dots \\ \hline 0 \dots \end{array}$$

or

$$\begin{array}{r} 0 \dots - \\ 1 \dots \\ \hline 1 \dots \end{array}$$

OF = 1.

$$\begin{array}{r} 0110 \ 0010 - \\ 1100 \ 1000 \\ \hline 1001 \ 1110 \end{array}$$

unsigned
CF = 1.

$$\begin{array}{r} 98 - \\ 200 \\ \hline -102 \end{array}$$

$$\begin{array}{r} 62h \\ c8h \\ \hline 9Ah \end{array}$$

$$\begin{array}{r} 98 - \\ -51 \\ \hline 15h \\ \text{signed.} \end{array}$$